

Giseop KIM

Assistant Professor | Principal Investigator

Department of Robotics and Mechatronics Engineering, DGIST

Joint appointments: Department of Artificial Intelligence; Mechanical Engineering Track

Director, Autonomy and Perceptual Robotics Lab (APRL)

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Research mission. APRL advances spatial intelligence and robot world models to enable autonomous robots to navigate, reason, and learn from experience in the real world.

Research profile. Robotics researcher working at the intersection of simultaneous localization and mapping (SLAM), spatial and physical AI, 3D perception, sensor fusion, long-term autonomy, and visual-language robot navigation. Current research investigates memory-augmented spatial intelligence, shared robot experience, embodied reasoning, and robot world models for robust real-world navigation.

Academic Appointments and Professional Experience

Dec. 2024–present	Assistant Professor , Daegu Gyeongbuk Institute of Science and Technology (DGIST), Daegu, Republic of Korea. Principal Investigator, APRL.
Dec. 2021–Dec. 2024	Research Scientist , NAVER LABS. Research in autonomous driving, 3D vision, mapping, and localization.
Mar. 2017–Aug. 2021	Graduate Researcher , KAIST, Intelligent Robotic Autonomy and Perception Lab.

Education

2022	Ph.D., Civil and Environmental Engineering , KAIST. Advisors: Prof. Ayoun Kim and Prof. Youngchul Kim.
2019	M.S., Civil and Environmental Engineering , KAIST.
2017	B.S., Civil and Environmental Engineering , KAIST.

Research Directions

Spatial intelligence for navigation	SLAM 2.0 for the Robot Web era; neural map representations; human-robot interactive visual-language navigation; long-term and multi-robot autonomy.
Embodied reasoning and world models	Reasoning for robots; generative AI for mobile robot navigation; memory-augmented spatial intelligence; robot world models.

Funded Research

Ongoing Projects

- **Robot Industry Technology Development Program** (Co-Principal Investigator)
 - **Period:** Jul. 2026–Dec. 2029 **Sponsor:** Ministry of Trade, Industry and Energy
 - **Program:** 2026 Robot Industry Technology Development Program (Core Robot Industry Technology Development)
 - **Title:** Spatial-RFM: Development of a VLM-based Spatial Intelligence Robot Foundation Model for Humanoid and Quadruped Robot Navigation in Unstructured Environments
- **AIMS: Memory-Augmented Spatial Intelligence** (Principal Investigator)
 - **Period:** Mar. 2026–Feb. 2030 **Sponsor:** Ministry of Science and ICT
 - **Program:** NRF Early-career Research Program (Type B), Basic Research Program 2026
 - **Title:** Memory-Augmented Spatial Intelligence for Heterogeneous Autonomous Mobility Systems Understanding and Interacting with Implicit Human Instructions
- **Industry Project 2026-1 (Confidential)** (Principal Investigator)
 - **Period:** Apr. 2026–Mar. 2027 **Sponsor:** Industry-funded
 - **Program:** Industry-academic technical service
 - **Title:** Mapping and Localization in Factory Environments
- **Basic Research Laboratory, Exploratory Type** (Co-Principal Investigator)
 - **Period:** Jul. 2026–Jun. 2029 **Sponsor:** Ministry of Science and ICT
 - **Program:** Basic Research Program, Basic Research Laboratory (Exploratory Type) 2026
 - **Topic:** Fundamental Study on Morphogenetic Cooperative Mobility Intelligence based on Physical Agentic

AI

- **Glocal Lab** (Participating Researcher)
 - **Period:** Sep. 2025–Aug. 2028 (+6 years) **Sponsor:** Ministry of Education
 - **Program:** Glocal Lab Program, Academic Research Support Program for Science and Engineering 2025
 - **Center:** TransHuman Robotics Research Center, Robotics and Mechatronics Research Institute of DGIST
- **InnoCORE** (Participating Researcher)
 - **Period:** Jul. 2025–Dec. 2027 (+3 years) **Sponsor:** Ministry of Science and ICT
 - **Title:** DGIST Team on Bio-Embodied Physical AI
- **AI Star Fellowship** (Participating Researcher)
 - **Period:** 2025–2030 **Sponsor:** Ministry of Science and ICT
 - **Title:** Human-centered Symbiotic Embodied AI Systems
- **DGIST Start-up Research** (Principal Investigator)
 - **Period:** Jan. 2025–Dec. 2028 **Sponsor:** DGIST
 - **Title:** Self-evolving Robot-Web Navigation Intelligence

Completed Projects

- **P2P Autonomous Driving Platform** (Participating Researcher)
 - **Period:** Sep. 2025–Mar. 2026 **Sponsor:** Ministry of Trade, Industry and Energy
 - **Program:** Autonomous Driving Technology Development Innovation Program, Korea Evaluation Institute of Industrial Technology
 - **Title:** Development of a Level-4 Passenger-Van Autonomous Driving Platform for Point-to-Point Mobility in Designated Areas
- **N-HRHR** (Participating Researcher)
 - **Period:** Jul.–Dec. 2025 **Sponsor:** DGIST
 - **Title:** Exploring Human-Robot Collaborative Coexistence in the Era of Large-Scale Robots

Honors and Awards

Sep. 30, 2026	Best Paper Award , IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026), for “LT-Mem: Volatility-Aware Spatio-Temporal Memory for Life-long Scene Understanding.” Recipients: Yumin Lee, Hyoseok Ju, and Giseop KIM.
Jul. 2026	Outstanding Young Researcher Award , 41st Annual Conference of the Institute of Control, Robotics and Systems (ICROS 2026).
Feb. 2026	Early-career Researcher Award , Korea Robotics Society Annual Conference (KRoC 2026).
Oct. 2025	Best Poster Award , Human-aware Embodied AI Workshop at the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025) (student advisee: Jiseon Kim).

Academic Leadership and Service

2027	Workshops & Tutorials Committee Co-Chair , IEEE International Conference on Robotics and Automation (ICRA 2027). <i>Forthcoming</i> .
2026	Workshop Organizer , Second Workshop on Human-aware Embodied AI, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026).
2026	Workshop Organizer , Long-Term Perception for Human-Centric Autonomy: Towards Undo for the Physical World, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026).
2026	Organizing Committee Member , 41st Annual Conference of the Institute of Control, Robotics and Systems (ICROS 2026).
2026	Tutorial Organizer and Speaker , Spatial AI Tutorial, Korea Robotics Society Annual Conference (KRoC 2026).

Publications

► first/co-first or corresponding-author paper by Giseop KIM; *first/co-first author; †corresponding author. Names of the CV author are in bold. Accepted papers are marked “to appear.”

Peer-Reviewed Journal Articles

1. S. Lee*, H. Yu, **Giseop KIM**, and S. Choi†, “LAMP: Implicit Language Map for Robot Navigation,” *IEEE Robotics and Automation Letters*, vol. 10, no. 12, pp. 12365–12372, 2025.
2. M. Kim*, **Giseop KIM**, and S. Choi†, “Addressing Diverging Training Costs Using BEVRestore for High-Resolution Bird’s Eye View Map Construction,” *IEEE Robotics and Automation Letters*, 2024.
3. H. Kim*, J. Choi, T. Sim, **Giseop KIM**, and Y. Cho†, “Narrowing Your FOV with SOLiD: Spatially Organized and Lightweight Global Descriptor for FOV-Constrained LiDAR Place Recognition,” *IEEE Robotics and Automation Letters*, 2024.
4. M. Jung*, W. Yang, D. Lee, H. Gil, **Giseop KIM**, and A. Kim†, “HeLiPR: Heterogeneous LiDAR Dataset for Inter-LiDAR Place Recognition under Spatiotemporal Variations,” *The International Journal of Robotics Research*, vol. 43, no. 12, pp. 1867–1883, 2024.
5. ► **Giseop KIM***, S. Choi, and A. Kim†, “Scan Context++: Structural Place Recognition Robust to Rotation and Lateral Variations in Urban Environments,” *IEEE Transactions on Robotics*, 2022.
6. Y. Cho*, **Giseop KIM**, S. Lee, and J.-H. Ryu†, “OpenStreetMap-based LiDAR Global Localization in Urban Environment without a Prior LiDAR Map,” *IEEE Robotics and Automation Letters*, 2022.
7. ► **Giseop KIM***, B. Park, and A. Kim†, “1-Day Learning, 1-Year Localization: Long-term LiDAR Localization using Scan Context Image,” *IEEE Robotics and Automation Letters*, 2019.

Peer-Reviewed International Conference Papers

8. ► H. Kim*, B. Kim, and **Giseop KIM**†, “MarsLab: A Martian Rover Simulator for Planetary Rover Autonomous Navigation,” *International Conference on Space Robotics (iSpaRo)*, 2026, accepted, to appear.
9. ► H. Kim* and **Giseop KIM**†, “Simulation for Planetary Robotic Perception and Autonomy: A Concise Survey of Recent Capabilities and Gaps,” *iSpaRo*, 2026, accepted, to appear.
10. ► Y. Lee*, H. Ju, and **Giseop KIM**†, “LT-Mem: Volatility-Aware Spatio-Temporal Memory for Lifelong Scene Understanding,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. **Best Paper Award**.
11. ► B. Nayak*, H. Ju*, and **Giseop KIM**†, “Mag4D-SLAM Dataset: A Repeated-Traversal Multi-Modal 4D Geomagnetic Dataset for Localization and Mapping,” *IROS*, 2026.
12. S. Moon*, J. Baek, Y. Jeong, D. Chae, **Giseop KIM**, J. Lee, J. Kim†, and S. Choi†, “Streaming Dense Voxel Representations for 3D Occupancy Prediction,” *European Conference on Computer Vision (ECCV)*, 2026.
13. ► B. Suh*, H. Ju, and **Giseop KIM**†, “What Matters for Real-World Long-Horizon Robot Navigation?: An Experimental Study of Implicit Goals and Sparse Memory,” *23rd IFAC World Congress*, 2026.
14. J. Lee*, J. Kim*, W. Shin, **Giseop KIM**, and H. Oh†, “Learning Point Cloud Geometry as a Statistical Manifold: Theory and Practice,” *Robotics: Science and Systems (RSS)*, 2026.
15. ► H. Ju*, B. Suh, and **Giseop KIM**†, “Have We Mastered Scale in Deep Monocular Visual SLAM? The Scale-Master Dataset and Benchmark,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
16. M. Kim*, D. Lee, J. Yu, J. Hur, **Giseop KIM**, and J. Kim†, “Inlier-Centric Post-Training Quantization for Object Detection Models,” *International Conference on Learning Representations (ICLR)*, 2026.
17. J. Kim*, S. Jeong*, **Giseop KIM**, M.-H. Jeon, E. Jun, and A. Kim†, “2D Gaussian Splatting-based Sparse-view Transparent Object Depth Reconstruction via Physics Simulation for Scene Update,” *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.
18. H. Gil*, D. Lee*, **Giseop KIM**, and A. Kim†, “Ephemerality Meets LiDAR-based Lifelong Mapping,” *ICRA*, 2025.
19. M. Kim*, **Giseop KIM**, K.-H. Jin, and S. Choi†, “BroadBEV: Collaborative LiDAR-Camera Fusion for Broad-Sighted Bird’s Eye View Map Construction,” *ICRA*, 2024.
20. H. Lim*, K. Han, G. Shin, **Giseop KIM**, S. Hong, and H. Myung†, “ORORA: Outlier-Robust Radar Odometry,” *ICRA*, 2023.
21. S. Yun*, M. Jung, J. Kim, S. Jung, Y. Cho, M.-H. Jeon, **Giseop KIM**, and A. Kim†, “STheReO: Stereo Thermal Dataset for Research in Odometry and Mapping,” *IROS*, 2022.
22. ► **Giseop KIM*** and A. Kim†, “LT-mapper: A Modular Framework for LiDAR-based Lifelong Mapping,” *ICRA*, 2022.
23. ► **Giseop KIM*** and A. Kim†, “Remove, then Revert: Static Point Cloud Map Construction using Multiresolution Range Images,” *IROS*, 2020.
24. ► **Giseop KIM***, Y. S. Park, Y. Cho, J. Jeong, and A. Kim†, “MulRan: Multimodal Range Dataset for Urban Place Recognition,” *ICRA*, 2020.
25. Y. Cho*, **Giseop KIM**, and A. Kim†, “Unsupervised Geometry-aware Deep LiDAR Odometry,” *ICRA*, 2020.
26. ► **Giseop KIM*** and A. Kim†, “Scan Context: Egocentric Spatial Descriptor for Place Recognition within 3D Point Cloud Map,” *IROS*, 2018.

Book Chapters

27. J. Behley, M. Fallon, S. Zhao, **Giseop KIM**, J. Zhang, F. Zhang, and A. Kim, “LiDAR SLAM,” Chapter 8 in *SLAM Handbook: From Localization and Mapping to Spatial Intelligence*, 2026.

Domestic Journal Articles

28. ▶ B. Suh^{*}, J. Kim, J. Son, M. Park, and **Giseop KIM[†]**, “Knowledge Fusion Strategies for Object Goal Navigation in the VLM Era: A Survey,” *Transactions of the Korean Society of Automotive Engineers*, vol. 34, no. 1, pp. 119–131, 2026.

Domestic Conference Papers

29. ▶ A. Lee^{*} and **Giseop KIM[†]**, “Floorplan-based Point-to-Language Grounded Navigation,” Institute of Control, Robotics and Systems Annual Conference (ICROS), 2026.
30. ▶ A. Lee^{*}, H. Yoon^{*}, S. Park^{*}, M. Lee^{*}, and **Giseop KIM[†]**, “Rendezvous Briefing: VLM-based Semantic-Aware Multi-Robot Collaborative Exploration,” ICROS, 2026.
31. ▶ **Giseop KIM^{*†}**, M. Park, and J. Son, “Change-Robust LiDAR Localization in Urban Environments,” Institute of Electronics and Information Engineers Conference, 2025.

Invited Talks and Tutorials

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| Sep. 27, 2026 | “From Maps to Memory: Bridging the Perception–Reasoning Gap for Long-Term Robot Navigation,” Workshop on Bridging Navigation, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026). |
| Sep. 4, 2026 | “Lidar Inertial SLAM for Spaceship Docking,” Invited Seminar, Korea Aerospace Research Institute (KARI). |
| Aug. 19, 2026 | “From Maps to Memory: Bridging Perception and Reasoning for Long-Term Autonomous Navigation,” Invited Seminar, Korea Institute for Robot Industry Advancement (KIRIA), Seoul. |
| Jul. 30, 2026 | “Geometric 3D Vision Made Easy? – The DUST3R Family and Its Evolution,” Invited Seminar, Electronics and Telecommunications Research Institute (ETRI), Daegu-Gyeongbuk Regional Research Division. |
| Jul. 24, 2026 | “Bridging the Perception–Reasoning Gap for Long-Term Robot Navigation (Feat. the Research Journey),” Invited Talk, Korea Advanced Institute of Science and Technology (KAIST). |
| Jul. 3, 2026 | “Visual Language Multi-robot SLAM,” Early-career Researcher Award Session, 41st Annual Conference of the Institute of Control, Robotics and Systems (ICROS 2026), Daegu. |
| Jun. 2, 2026 | “What, When, and Why Visual-Language Navigation?,” Private Meetup at the IEEE International Conference on Robotics and Automation (ICRA 2026), URobotics. |
| May 28, 2026 | “3D Perception, Language, and AI for Human-Robot Interactive Navigation,” Ewha Womans University. |
| Apr. 2, 2026 | “Memory-Augmented Spatial Intelligence for Autonomous Robot Navigation,” Hanyang University ERICA. |
| Mar. 24, 2026 | “Memory-Augmented Spatial Intelligence for Autonomous Robot Navigation,” ETRI Daegu-Gyeongbuk Regional Research Division. |
| Feb. 6, 2026 | “IMU Basics and Inertial Aided Navigation,” Spatial AI Tutorial Organizer and Speaker, Korea Robotics Society Annual Conference (KRoC 2026), Pyeongchang. |
| Feb. 5, 2026 | “Memory-Augmented Spatial Intelligence for Autonomous Robot Navigation,” Early-career Researcher Award Session, KRoC 2026, Pyeongchang. |
| Feb. 5, 2026 | “CoT, Scene Graphs, and Recent Paradigms for Mobile Robot Navigation: A Short Survey and Real-World Lessons,” Special Session on AI-based Autonomous Systems, KRoC 2026, Pyeongchang. |
| Feb. 2, 2026 | “Daegu’s Innovation, Moving Towards a Robot and Future Mobility City,” Dalseo-gu Office Monthly Meeting Special Lecture, Daegu. |
| Jan. 19, 2026 | “An Invitation to 3D Vision, Spatial AI, and Physical AI,” Invited Tutorial, International Conference on Electronics, Information, and Communication (ICEIC 2026), Macau. |
| Dec. 10, 2025 | “SLAM, Spatial AI, and Human-Robot Interactive Navigation,” Department of Intelligent Robotics, Sungkyunkwan University, Suwon. |
| Dec. 9, 2025 | “Robotics and Spatial AI for Construction,” Department of Civil and Environmental Engineering, KAIST, Daejeon. |

Dec. 8, 2025	“SLAM, Spatial AI, and Human-Robot Interactive Navigation,” Department of Industrial Machinery DX, Korea Institute of Machinery and Materials, Daejeon.
Oct. 30, 2025	“Physical AI for Smart Factory: Humanoid, Sensing, and Spatial Intelligence,” Fall Conference of the Korean Fiber Society, Busan.
Aug. 15, 2025	“Generative AI for Mobile Robot Navigation,” AI Robotics KR Conference: Human and AI Robotics Together, Seoul.
Jul. 18, 2025	“Generative AI for Mobile Robot Navigation,” Joint Summer School on Image Understanding and Image Processing, Institute of Electronics and Information Engineers, Seoul.
Jun. 13, 2025	“Lessons from Translating Research into Products and Services,” SLAM and Perception Seminar, Hyundai Motor Company Uiwang Research Institute.
May 28, 2025	“Spatial AI, from the 2000s to 2025: The Evolution of Spatial Intelligence and the Future of Robotics,” ETRI Daegu-Gyeongbuk Regional Research Division Mid- to Long-term Research Roadmap Meeting, Daegu.

Graduate Student Advising

Bokeon Suh	Integrated M.S./Ph.D.	Fall 2025–present
Jiseon Kim	M.S.	Fall 2025–present
Yumin Lee	M.S.	Fall 2025–present
Hyoseok Ju	M.S.	Fall 2025–present
Doyeon Kim	Ph.D.	Spring 2026–present
Beomsu Kim	M.S.	Spring 2026–present
Hoyun Kim	M.S.	Spring 2026–present

Teaching at DGIST

2026 Spring	RT604	Advanced Mobile System
2026 Spring	MECH301	Robots for Human
2025 Fall	MECH307	Introduction to Artificial Intelligence
2025 Fall	AT603	Introduction to Mobility Engineering
2025 Spring	BE203	Creative Mechanical Design

Research Group

Autonomy and Perceptual Robotics Lab (APRL), DGIST. Founded December 2024. Current full-time group: one postdoctoral researcher, two doctoral researchers (including one integrated M.S./Ph.D. student), and five M.S. students, with undergraduate interns and external collaborators across robotics, perception, and AI.